



Molecular and Microbiological Approaches to Address Western Sword Fern Decline Phase 1 Grant Report

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Abstract

In the last decade, there has been a rapid decline of western sword fern (*Polystichum munitum*) throughout the Puget Sound region. Preliminary data suggested an infectious agent as the cause, however, results were not definitive. We propose to use molecular and microbiological approaches to determine the probable cause of the mortality. Phase 1 objectives are to verify that inoculum from affected plants causes decline of healthy fern, and subsequently, to analyze the inoculum using PCR, sequencing, microscopy, and microbiological screening techniques to determine the pathogen identity. For a subsequent Phase 2 grant, we will propose to use plant microbial mutualists for biocontrol and increased resilience of western sword fern plants. Our current suite of microbial strains with strong activity against *Rhizoctonia* and other fungal pathogens will be tested for activity against the fern pathogen. The ability of the biocontrol strains to colonize and persist in fern will be ascertained. In parallel, the microbiota of fern naturally surviving in the decline zones will be screened for activity, purified, and identified. The overall aim of the full project is to provide a natural protective inoculum for use in restoration of western sword fern.

Phase 1: Determination of the cause of the sword fern decline

Phase 1 Research Plan:

- 1) Collect frond and rhizome samples from ferns in and near the affected zone as well as in unaffected zones
- 2) Confirm the previous result that the frond rinsate (inoculum) is infectious on healthy ferns. We will also test rhizome rinsates
- 3) Treat the inocula to determine if the causal agent is living, and if it is bacterial, fungal, or viral
- 4) Analyze the inocula for suspected pathogens through growth assays, microscopy, PCR, and sequencing
- 5) Submit report on the probable causal agent of the sword fern decline, and propose a Phase 2 research plan to investigate biocontrol strategies using natural PNW microbes

Molecular and Microbiological Approaches to Address Western Sword Fern Decline

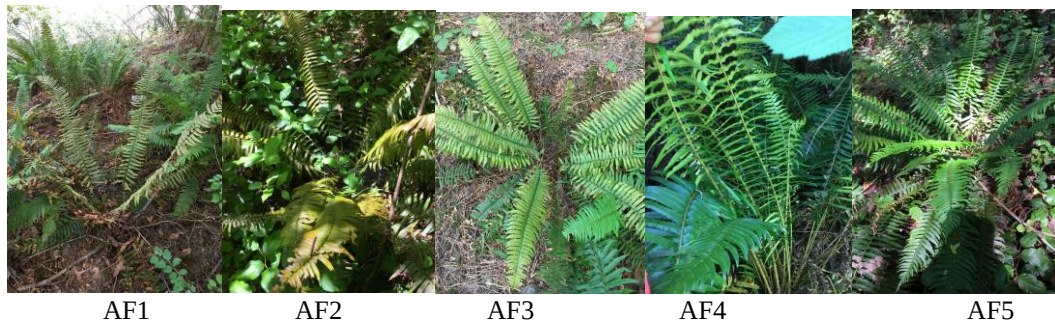
Phase 1 Grant Report

Experiment 1: Inoculation of healthy fern plants with infusions and extracts of affected ferns from Seward Park

Rationale: If fern decline is caused by a pathogenic agent, samples of affected ferns added to healthy ferns should result in disease. Completely dead ferns might not still contain the putative pathogen so we chose to use moderately affected ferns that appeared to be affected and dying. Monitoring disease progression would also allow for insights into disease pathology and inform downstream tissue sampling and microscopy.

Methods: Doty collected healthy fern plants from a semi-isolated site near the Skykomish River to serve as the healthy recipient ferns. Lisa Ciecko collected five moderately-affected ferns from Seward Park near the “ground zero” area in June 2020 according to our instructions (**Figure 1**). Sher and Doty processed the affected fern fronds and rhizomes in three ways from each frond used. Secreted fluids from cut fronds, water rinses of the samples, and extracts from

Figure 1. The five affected ferns that were collected by Lisa Ciecko for Experiment 1.



homogenized tissues were all collected. A portion of these samples was filtered through a 0.2 micron filters to remove bacterial and fungal cells; this sub-sample would retain only viruses. As a negative control, another sub-sample was sterilized by autoclaving (121C, 30 min). For each of the sub-samples and the original samples, three healthy ferns were inoculated in two ways. Treatments were poured directly onto the rhizome and also dabbed onto one frond of each fern (**Figure 2**). The three sets of ferns were separated by plastic sheeting and the experiment was isolated using a temporary barrier (**Figure 3**). Additionally samples from each of the 5 affected ferns were plated onto rich medium to visually assess microbial diversity.

Experiment 1 Results: After approximately one week, all the plants exhibited some stress including browning of frond tips, likely due to the lower humidity and higher light levels of our plant growth room compared to the original habitat of these ferns. Light levels were reduced, and trays of water were added to increase humidity levels in addition to daily misting of the ferns. After two weeks into the experiment, all the fronds of one plant in the filtered inoculum group and one in the live inoculum group had leaflette curling. However, both ferns began recovering after a few weeks, growing new fronds. There were no health differences between fronds that were directly inoculated through dabbing compared to those that were not. One month after the inoculation, the experiment was concluded with no fern mortalities (**Figure 4**).

Figure 2. Applying inoculum to test plants in Experiment 1. The solutions prepared from the five affected ferns were combined and divided equally between the three plants of each treatment. One frond per plant was directly inoculated using sterile cotton soaked in the inoculum. The remainder was poured into the soil at the base of each plant.



Figure 3. Treatment groups at the beginning of Experiment 1. A, received live inoculum; B, received filtered inoculum; C, received autoclaved inoculum

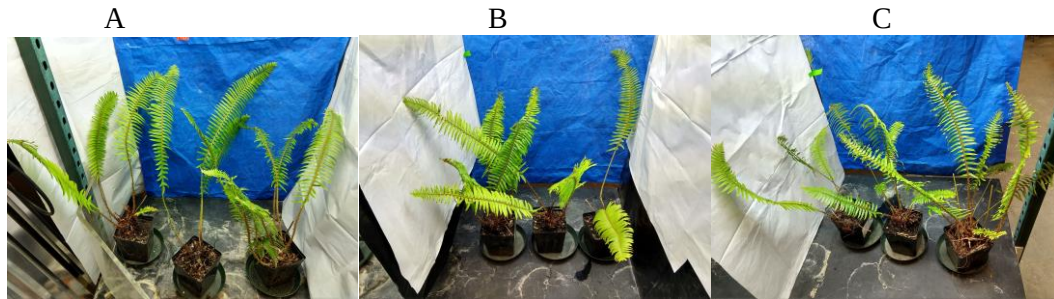
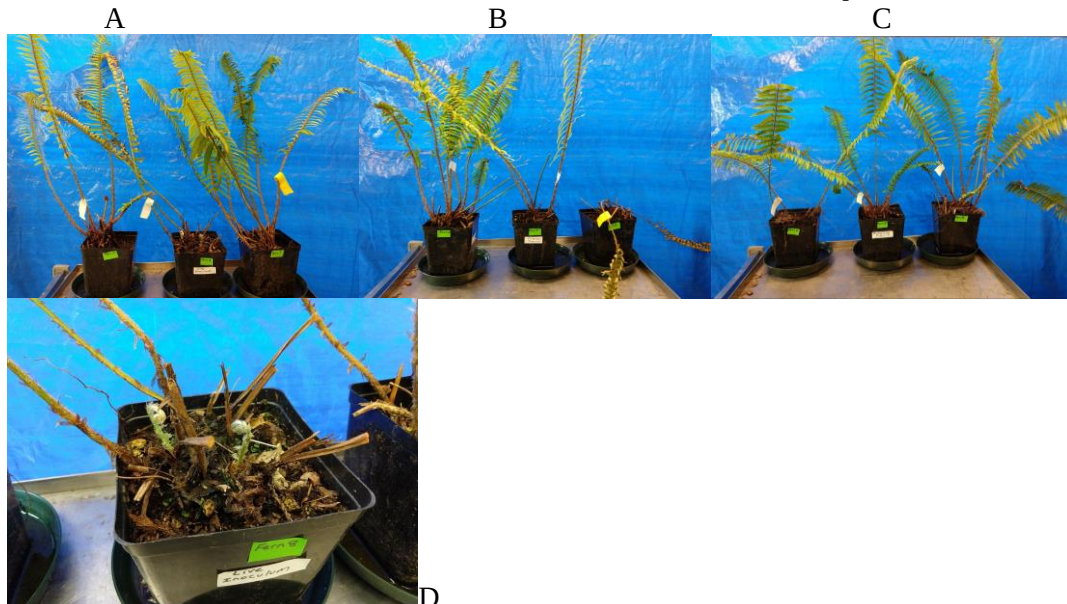


Figure 4. Experiment 1 treatment groups after 4 weeks. A, received live inoculum; B, received filtered inoculum; C, received autoclaved inoculum; D, new fronds developing on live-inoculum-inoculated fern # 8. Plants were retained for a total of 8 weeks with no development of disease.



Experiment 1 Conclusions: Despite direct addition of affected fern extracts, none of the inoculated ferns died. Only two ferns from two treatment groups experienced any frond loss but these regrew new fronds. Though there was some frond drying and loss, overall there were no significant differences in health between ferns that received autoclaved (sterilized) extract compared to those that received live extracts. Since even the two experimental plants that suffered severe stress recovered and had new growth, it can be concluded that there were no virulent pathogens in the inoculum. It is possible that the collected ferns were not truly affected so we address this concern in Experiment 3. Since all the ferns initially developed symptoms similar to those at Seward Park but recovered when light levels were lowered and humidity increased, we investigated the hypothesis that these causes were a factor as a prelude in Experiment 3.

Experiment 2: Inoculation of cut fronds from healthy plants

Rationale: Our original proposal was based on previous data from other groups that determined that cut fronds were more severely impacted by sharing water with an affected frond than with a healthy frond within 5 days. If the cause of the frond decline was biological, then exposure to the microorganisms found in the exudate of the affected fern fronds should result in pathogenic symptoms. Though this was also tested in Experiment 1, Experiment 2 sought to repeat previous results.

Experiment 2 Methods: Fronds from healthy plants were collected from the UW campus and immediately placed in water and taken to lab where they were divided in triplicate into conical tubes containing 30 ml of either an inoculum from affected fern fronds, autoclaved inoculum, blank solution, or bacterial strain WP5. The secretions from the five affected fern fronds in Experiment 1 had previously been cryogenically stored. These were streaked onto rich medium (MG/L), and colonies from each of the 5 plates were used to inoculate 100 ml MG/L broth and grown overnight. Half the culture was then autoclaved to serve as a negative control. Optical density of the live culture was measured using a spectrophotometer, and the culture adjusted to an OD₆₀₀ of 0.1 in 100ml water. For comparison, 100 ml water containing the same volume of MG/L broth was made as a blank control. As a microbial control, 100 ml of a mutualistic bacterial strain (WP5) was prepared in the same way. Fronds were misted daily with water and grown under low light for five days.

Experiment 2 Results: Within three days, all the fronds exhibited leaf curling and were essentially dead within one week (**Figure 5**). After 3 weeks when the experiment was terminated, it was noted that the average liquid uptake was the same across all treatment groups.

Experiment 2 Conclusions: The cut frond assay does not seem to be a valid assay

Figure 5. Frond assay results. A, beginning of experiment; B, after 5 days. Treatment groups from left to right: WP5, autoclaved inoculum, live inoculum, and blank solution

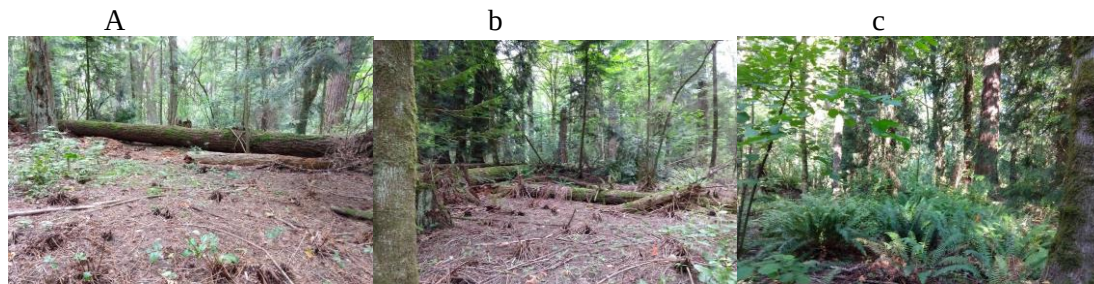


Experiment 3: Inoculation of healthy fern plants with homogenized severely affected fern from Seward Park

Rationale: If the ferns collected from Seward Park for Experiment 1 were not diseased sufficiently long enough to harbor a large pathogenic load, we chose to sample a fern plant with severe decline symptoms. The plant was selected from an area where other potential causes of fern decline were not present.

Investigations of sites of sword fern decline and causes: Examination of several sites of sword fern decline within Seward Park revealed that most cases could likely be attributed to environmental factors. The “ground zero” area had lost much of its evergreen canopy coverage due to laminated root rot and other root rot diseases of the mature Douglas-fir trees. Much of the site now receives full sun and the soil was dry. An ecophysiology study ¹showed that the ferns at the site had Fv/Fm values at all sampling times in 2018 and 2019 that were below 0.80 which is indicative of stress for ferns ². Accompanying vegetation at each site of sword fern decline compared to sites with healthy ferns in Seward Park provided more evidence of dry conditions. For example, Oregon grape, dead beaked hazelnut, dying salal and oceanspray, and sparsely foliated western hemlock were common co-inhabitants of the declining fern sites whereas red huckleberry, thimbleberry, salmonberry, and conifers were present with the healthy ferns (**Figure 6**). In nearly all of the sites of sword fern decline, there were downed mature conifers

Figure 6. Seward park sword fern decline sites (a and b) and healthy fern sites (c). Dead ferns are co-incident with downed mature trees.

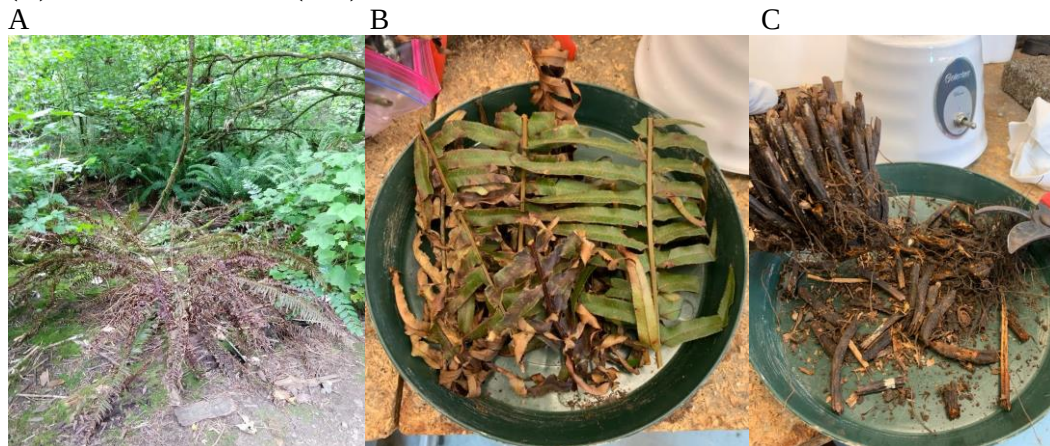


and an open canopy or dominated by deciduous trees resulting in a far less shaded understory. Ferns planted into the “ground zero” site that have been watered appeared healthy. One fern decline site that did not have downed mature conifers was instead severely impacted by mountain beavers as evidenced by a multitude of large holes near the dying ferns. Though it is thought that the beavers feed on fern fronds, judging by the holes, it may be that they also feed on the fern rhizomes. Sites of sword fern decline at the entrance to Japanese Gardens at the Washington Park Arboretum and at the Lord Hill Regional Park were also viewed. In each area of dying sword fern, evidence of changes in the overstory such as fallen branches and dying trees and dryness-adapted plant species accompanied the declining fern. Adjacent areas of healthy fern were not exposed to full sun and were mixed with moisture-adapted plant species including horsetail ferns and huckleberry.

Experiment 3 Methods: One severely affected fern at Seward Park that was not near beaver holes or in a dry environment was selected as an inoculum source for Experiment 3. The fern was flanked by healthy thimbleberry and salmonberry and was within a shaded understory (**Figure 7**). Fronds and soil were collected and brought to lab where it was immediately processed in a blender with water. Since the ferns collected for Experiment 1 had exhibited stress symptoms from the separation process into small pots, a fresh set of healthy Skykomish River area ferns were collected and placed into large pots with their native soil, and were kept

moist throughout the experiment. Five of the ferns received the blended inoculum. Pieces of the affected fern were also placed at the base of each the plants. Another set of five ferns received no treatment. All the plants were well-watered and misted daily.

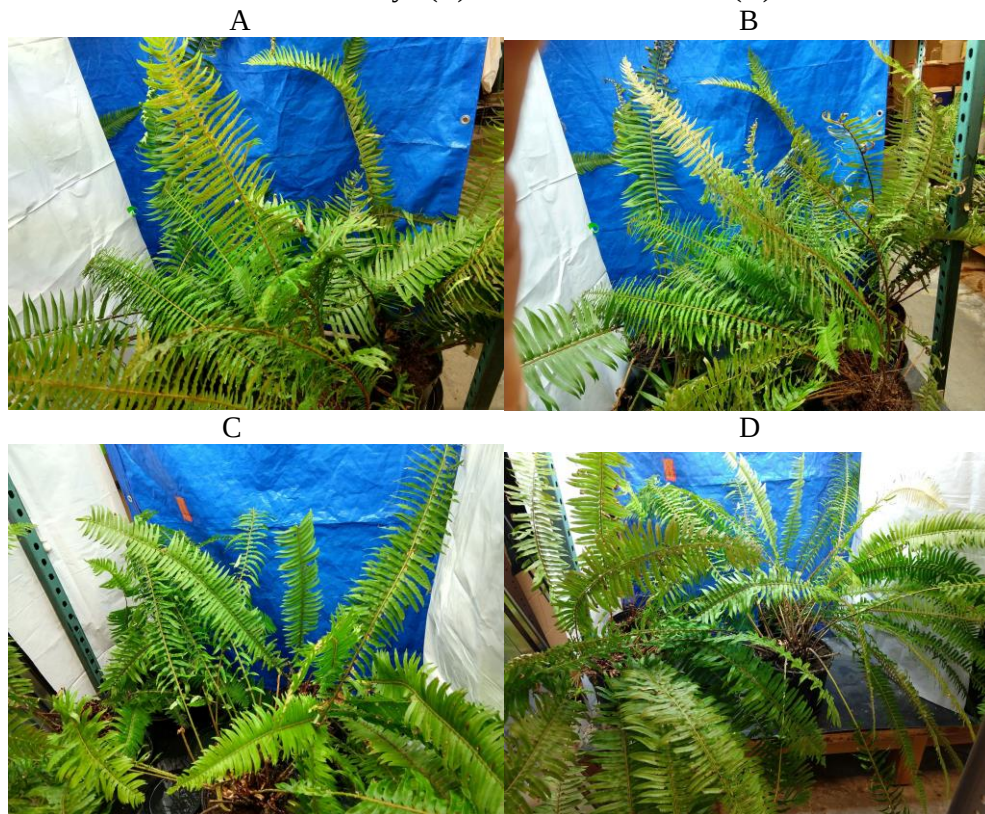
Figure 7. Unhealthy fern at Seward Park selected as the inoculum source for Experiment 3 (A). Frond and rhizome (B,C) were cut and blended with soil to create the inoculum.



Experiment 3 Results:

All of the plants remained healthy for the duration of the three week study (Figure 8).

Figure 8. Experiment 3 uninoculated ferns at Day1 (A) and after three weeks (B); inoculated ferns at Day1 (C) and after three weeks (D)



Discussion

The results from the three experiments did not support the hypothesis that fern decline is caused by a transmissible agent. The majority of fern plants in Experiments 1 and 3 remained healthy whether or not they received inoculum from the Seward Park fern plants. The two ferns in the first experiment that experienced frond loss quickly grew new fronds. Experiment 1 suffered from low humidity, and signs of plant stress were seen in all treatment groups. In Experiment 3, this was corrected with daily misting, and no treatment groups experienced frond loss or even showed signs of stress. The results of Experiment 2 indicated that the cut frond assay is not a reliable test since all fronds died within a week, regardless of treatment group and despite misting.

Analysis of the sword fern decline sites, as well as our own experience in transplanting healthy ferns from an intact forest into a plant growth room, suggested that dry conditions cause sword fern decline. The Pacific Northwest has been experiencing drought conditions coincident with the period of fern decline. In the West, 2011 to 2015 was marked with a longer duration of warm, dry weather, and across the Northwest, the year 2015 was listed as an extreme drought year with exceptionally low snowpack and higher temperatures^{3,4}. At each of the sites of sword fern decline that we investigated, several Pacific NW iconic species were also declining, again suggesting that a specific pathogen is not the cause of the decline. Projected climate changes affecting western USA include an increased frequency of severe drought, reduced snowpack, and hotter and drier growing seasons⁵.

Though no evidence for a pathogen was observed in this study, the possibility remains that a pathogen was responsible for some of the die back observed at Seward Park. If a pathogen had moved through the area, it may have already killed those plants most susceptible and currently only remains in the environment in small numbers. Fern plants that others had transplanted into the die-back sites and were irrigated appear to be doing well. Without active spread and infection, identification of a pathogen is unlikely. The die back observed may also be the result of a combination of factors, with environmental stress triggering disease outbreak. Though these possibilities exist, it seems unlikely that a novel or virulent pathogen is responsible for the die back seen in Seward Park.

Budget: Since there was no detected pathogen to identify, the funds budgeted for molecular identification of the microorganism were not used and will therefore be returned.

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